

Reviewer Pack — Minimal Rank of Geometric Group Actions vs AC0 Parity Circui...

Entry 168090a28cba · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Geometric Group Actions vs AC0 Parity Circuit Threshold

Entry ID: 168090a28cba **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 00:17:55 UTC

1. Conjecture

Field A (mathematical branch): Geometric Group Theory **Field B** (complexity object): Complexity Theory: AC0 Parity Circuit Complexity

Statement:

[‘For every geometric group G , there exists a constant c_G such that any AC0 parity circuit C computing XOR on n inputs with size $S(C)$ has a minimal rank of G -rank($G(S(C)) \leq c_G \cdot \log(S(C))$).’, ‘If this is not the case for some group G and constant c_G , then there exists an AC0 parity circuit C computing XOR on n inputs with size $S(C)$ such that $r(G(S(C))) > c_G \cdot \log(S(C))$, where $r(G)$ denotes the $r(G)$ -invariant subset of a topological space.’, ‘For any fixed G , as n increases, this ratio converges to a finite limit or diverges.’]

Rationale (proposer’s reasoning):

[‘Geometric group theory studies groups that are finitely presented and act on geometric spaces. These groups can be used to encode information about the structure of the space they act on.’, ‘By considering the action of these groups on circuits, we may gain insight into the complexity of computing parity on those circuits.’, ‘A connection between the rank of a group action and circuit complexity could reveal new structural properties of complexity classes.’]

Taxonomy category: KARCHMER_WIGDERSON (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0d9b44588638e3bf

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if for all geometric groups G , at least 80% of the seeds yield a ratio $r(G(S(C))) \leq c_G \cdot \log(S(C))$ where $S(C)$ is the size of the AC0 parity circuit and c_G is the group-specific constant. The conjecture is falsified if any seed produces a ratio greater than $c_G \cdot \log(S(C))$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 5 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): -geometric group theory AND AC0 parity circuit complexity - AC0 parity circuit threshold IN geometric group theory - $r(G)$ -invariant subset IN complexity theory AND $G\text{-rank}(G(S(C))) \leq c_G \cdot \log(S(C))$

Top relevant hits considered: - [http://arxiv.org/abs/1611.00827v2] Geometric complexity theory and matrix powering - [http://arxiv.org/abs/2309.16111v2] The relational complexity of linear groups acting on subspaces - [http://arxiv.org/abs/2510.10838v1] Survey on Cremona groups from a median geometric point of view - [http://arxiv.org/abs/2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [http://arxiv.org/abs/hep-ph/0610012v1] Tevatron-for-LHC Report of the QCD Working Group

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=2.4s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Generate a random geometric group G (simplified for demonstration)
    n = 5 + random.randint(0, 30)
    G = {i: set(range(n)) for i in range(n)}

    # Construct an AC0 parity circuit C with increasing sizes
    S_C_values = [10**i for i in range(1, 6)]
    ratios = []

    for S_C in S_C_values:
        # Compute the r(G)-invariant subset for each group action on the circuit
        r_G_S_C = sum(len(G[i] & set(range(S_C))) for i in G) / len(G)

        # Compare it to c_G * log(S(C))
        if n == 0: continue # Avoid division by zero
        c_G = math.log(n) / S_C
        ratio = r_G_S_C / (c_G * math.log(S_C))
```

```

        ratios.append(ratio)

    # Measure the rank of the invariant subset and compare it to  $c_G \cdot \log(S(C))$ 
    metric_value = sum(ratios) / len(ratios)
    conjecture_holds = all(r <= 1 for r in ratios)
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "Rank Ratio",
        "metric_value": metric_value,
        "instances_tested": len(S_C_values),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2**i + 3 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'metric_value': 6658.783286344985, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 19166.447133957437, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 17534.212679479937, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 15439.928585244541, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 18761.630094680142, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 8635.900900505094, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 14582.699744989915, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 9602.026232172659, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'Rank Ratio', 'metric_value': 9121.548386637962, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f83278c5.py", line 70, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)

```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f83278c5.py", line 70, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                             ~^^^^^^^^^^
```

KeyError: 'seed'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that for all geometric groups G tested, the ratio $r(G(S(C))) > c \cdot G \cdot \log(S(C))$ was observed in at least one instance, which vi | next: Investigate the counterexamples provided to understand the nature of the violations and refine the conjecture or explore alternative approaches.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15208
2	propose	ollama_remote	glm4:latest	0	0	14874
3	propose	ollama_remote	glm4:latest	0	0	14409
4	preregistration	ollama_remote	glm4:latest	0	0	9502
5	novelty	ollama_remote	glm4:latest	0	0	8336
6	novelty	ollama_remote	glm4:latest	0	0	9435
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14757
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12178
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8385
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7565
11	judge	ollama_remote	glm4:latest	0	0	14559

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 129208 ms total latency. Provider mix: {'ollama remote': 11}

(full prompt+response transcripts available in [research/audit/168090a28cba.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/168090a28cba.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/168090a28cba.tar.gz (if generated)